

Task 1: Python File Handling & Automation

```
def write_text_file():

    try:

        filename = input("Enter TXT filename (example: notes.txt): ")

        # Taking user input for file content

        content = input("Enter content to write into the file: ")

        # Opening file in write mode

        with open(filename, "w") as file:

            file.write(content)

        print("TXT file created and data written successfully.\n")

    except Exception as e:

        print("Error while writing TXT file:", e)

# =====

# Function to read a TXT file

# =====

def read_text_file():

    try:

        filename = input("Enter TXT filename to read: ")

        # Opening file in read mode

        with open(filename, "r") as file:

            data = file.read()

        print("\nFile Content:")

        print(data)

        print()

    except FileNotFoundError:

        print("Error: File does not exist.\n")

    except Exception as e:

        print("Error while reading TXT file:", e)

# =====

# Function to create and write data into a CSV file

# =====

def write_csv_file():

    try:

        filename = input("Enter CSV filename (example: students.csv): ")

        # Opening CSV file in write mode

        with open(filename, "w", newline="") as file:
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writer = csv.writer(file)

# Writing heading row
writer.writerow(["ID", "Name", "Marks"])

# Number of records
n = int(input("Enter number of student records: "))

# Taking user input for records
for i in range(n):
    sid = input("Enter Student ID: ")
    name = input("Enter Student Name: ")
    marks = input("Enter Student Marks: ")
    writer.writerow([sid, name, marks])
    print("CSV file created successfully.\n")

except ValueError:
    print("Invalid input. Please enter numeric values properly.\n")

except Exception as e:
    print("Error while writing CSV file:", e)

# =====

# Function to read CSV file
# =====

def read_csv_file():
    try:
        filename = input("Enter CSV filename to read: ")

        # Opening CSV file in read mode
        with open(filename, "r") as file:
            reader = csv.reader(file)

            print("\nCSV File Data:")

            for row in reader:
                print(row)

            print()

    except FileNotFoundError:
        print("Error: CSV file not found.\n")

    except Exception as e:
        print("Error while reading CSV file:", e)

# =====

# Function to rename a file
# =====

def rename_file():

```

```

try:

old_name = input("Enter current filename: ")
new_name = input("Enter new filename: ")

# Renaming file

os.rename(old_name, new_name)

print("File renamed successfully.\n")

except FileNotFoundError:

print("Error: File not found.\n")

except Exception as e:

print("Error while renaming file:", e)

# =====

# Function to move a file

# =====

def move_file():

try:

source = input("Enter source filename/path: ")

destination = input("Enter destination folder/path: ")

# Moving file

shutil.move(source, destination)

print("File moved successfully.\n")

except FileNotFoundError:

print("Error: Source file not found.\n")

except Exception as e:

print("Error while moving file:", e)

# =====

# Function to delete a file

# =====

def delete_file():

try:

filename = input("Enter filename to delete: ")

# Deleting file

os.remove(filename)

print("File deleted successfully.\n")

except FileNotFoundError:

print("Error: File not found.\n")

except Exception as e:

print("Error while deleting file:", e)

```

```

# =====

# Main Menu Driven Program

# =====

while True:

    print("=====")

    print("    FILE HANDLING & AUTOMATION SYSTEM")

    print("=====")

    print("1. Write TXT File")

    print("2. Read TXT File")

    print("3. Write CSV File")

    print("4. Read CSV File")

    print("5. Rename File")

    print("6. Move File")

    print("7. Delete File")

    print("8. Exit")

    print("=====")

    # Taking user choice

    choice = input("Enter your choice: ")

    # Calling functions according to user choice

    if choice == "1":

        write_text_file()

    elif choice == "2":

        read_text_file()

    elif choice == "3":

        write_csv_file()

    elif choice == "4":

        read_csv_file()

    elif choice == "5":

        rename_file()

    elif choice == "6":

        move_file()

    elif choice == "7":

        delete_file()

    elif choice == "8":

        print("Program exited successfully.")

        break

    else:

```

```
print("Invalid choice. Please try again.\n")
```

```
=====
FILE HANDLING & AUTOMATION SYSTEM
=====
1. Write TXT File
2. Read TXT File
3. Write CSV File
4. Read CSV File
5. Rename File
6. Move File
7. Delete File
8. Exit
=====
Enter your choice: 1
Enter TXT filename (example: notes.txt): BOOK1.txt
Enter content to write into the file: Hi Everyone
TXT file created and data written successfully.

=====
FILE HANDLING & AUTOMATION SYSTEM
=====
1. Write TXT File
2. Read TXT File
3. Write CSV File
4. Read CSV File
5. Rename File
6. Move File
7. Delete File
8. Exit
=====
Enter your choice: 2
Enter TXT filename to read: BOOK1.txt
File Content:
Hi Everyone

=====
FILE HANDLING & AUTOMATION SYSTEM
=====
1. Write TXT File
2. Read TXT File
3. Write CSV File
4. Read CSV File
5. Rename File
6. Move File
7. Delete File
8. Exit
=====
Enter your choice: 3
Enter CSV filename (example: students.csv): students.csv
Enter number of student records: 2
Enter Student ID: 101
Enter Student Name: Adi
Enter Student Marks: 98
Enter Student ID: 102
Enter Student Name: Myra
Enter Student Marks: 75
CSV file created successfully.

=====
FILE HANDLING & AUTOMATION SYSTEM
=====
1. Write TXT File
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4. Read CSV File
5. Rename File
6. Move File
7. Delete File
8. Exit
```

```
=====
Enter your choice: 4
Enter CSV filename to read: students.csv
```

```
CSV File Data:
['ID', 'Name', 'Marks']
['101', 'Adi', '98']
['102', 'Myra', '75']
```

```
=====
FILE HANDLING & AUTOMATION SYSTEM
=====
```

1. Write TXT File
2. Read TXT File
3. Write CSV File
4. Read CSV File
5. Rename File
6. Move File
7. Delete File
8. Exit

```
=====
Enter your choice: 5
Enter current filename: Book1.txt
Enter new filename: Book1.txt
File renamed successfully.
```

```
=====
FILE HANDLING & AUTOMATION SYSTEM
=====
```

1. Write TXT File
2. Read TXT File
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4. Read CSV File
5. Rename File
6. Move File
7. Delete File
8. Exit

```
=====
Enter your choice: 6
Enter source filename/path: Book1.txt
Enter destination folder/path: Book1.txt
File moved successfully.
```

```
=====
FILE HANDLING & AUTOMATION SYSTEM
=====
```

1. Write TXT File
2. Read TXT File
3. Write CSV File
4. Read CSV File
5. Rename File
6. Move File
7. Delete File
8. Exit

```
=====
Enter your choice: 7
Enter filename to delete: Book1.txt
File deleted successfully.
```

```
=====
FILE HANDLING & AUTOMATION SYSTEM
=====
```

1. Write TXT File
2. Read TXT File
3. Write CSV File
4. Read CSV File
5. Rename File
6. Move File
7. Delete File
8. Exit

```
=====
Enter your choice: 8
Program exited successfully.
```

```
...Program finished with exit code 0
Press ENTER to exit console.
```